Directed Cyclic Networks
Phase 1: Literature and Novelty Audit

20 September 2026

Research audit, not a manuscript or a proof compendium. The six supplied papers were inspected. Additional primary sources provide methodological context; access gaps are explicit. All project proof modules remain TODO.

Contents

1 Literature audit: scope and evidence **2**

2 Six supplied primary sources **3**

3 Targeted methodological and five-element supplements **6**

4 Audit consequences and unresolved evidence **8**

5 Novelty matrix **10**

1 Literature audit: scope and evidence

1.1 Contract, corpus and limits

This audit addresses the eight novelty questions for the canonical demand-modulated model $(\mathrm{DM})_{n,q}$ and its matched bilinear comparator $(\mathrm{MB})_{n,q}$. Their equations and assumptions remain exactly those in `00_canonical_model.tex`. The main scope is arbitrary $n \geq 5$ and $1 \leq q \leq n-1$, with the exact selection program concentrated on $q = 2$. Five-element literature is integrated wherever it affects the model, coupling law, $n = 5, q = 2$ specialization or comparator; it is not a separate archival research track.

The six author-supplied PDFs form the core corpus, S01–S06. They were accessible at the local paths supplied on 20 September 2026, although the project's synced `sources/` directory still contained only the master blueprint. S07–S14 are targeted primary-source supplements or expressly unresolved leads. No master-blueprint statement is evidence of novelty. No project theorem has been derived, no experiment run, and no source's numerical findings independently reproduced.

Locator convention. For S01–S04, page numbers below are the *printed pages in the supplied author PDFs*, which also equal their one-based PDF positions. They are not inferred journal pages. S05 has printed pages 777–781 (PDF positions 1–5); S06 has publisher pages 1–15. The local filenames, page counts and SHA-256 digests are in `audits/phase1_source_manifest.md`. Supplemental versions and pagination are identified individually. A missing full-text locator is recorded as an access gap, never manufactured from an abstract.

Method. The audit compared each displayed source model with the canonical coupling locations, signs, state dependence, dimension and parameter restrictions. It inspected the relevant statements and calculations for spectra, mode selection, Hopf coefficients and perturbations, using text extraction and visual checks of critical equations. Exact titles and DOIs, circulant Hopf reductions, inverse-matrix decay, and five-element enzymatic antecedents were searched in publisher and author repositories. This is a bounded literature audit, not an exhaustive priority search. "Not located" is restricted to the inspected source; it does not establish nonexistence in the literature. Source numerical reports remain reports, even when placed under a theorem heading.

1.2 Novelty vocabulary

The following categories describe the *scope of a claim*, not its proof status in this project.

Definitely known (K).
The stated general fact, model component or specific antecedent is explicitly present in a checked source.

Directly implied by known theory (D).
Once the project's required hypotheses are independently verified, a known result supplies the claimed mechanism or representation. This is an audit inference, not a proof here.

Apparently new specialization (S).
No exact instance was located, but the proposed contribution applies established machinery to this particular model or parameter family. Priority remains provisional.

Apparently new theorem (T).
A precisely stated mathematical addition beyond a routine specialization appears absent from

the reviewed literature. This label requires an identifiable additional conclusion and adequate literature coverage; it does not mean the theorem is true.

Unresolved because more literature is needed (U).
Source access, the extent of existing theory or the exact proposed statement is insufficient for a defensible novelty verdict.

No complete project theorem receives an unconditional T label in this phase. Quantitative sign regions and parameter-dependent robustness certificates remain possible theorem contributions, but their exact scope and nearest antecedents are not sufficiently settled to award that label. The matrix distinguishes these prospects from established methods.

2 Six supplied primary sources

2.1 S01: Jaramillo, Mrad and Stepien (2023)

Source/version. [5]; supplied `2210.04342v2.pdf`, arXiv v2, 14 pages.

Exact model and assumptions. Section III, p. 3, Eqs. (5)–(6c) adds global linear mutation to a three-species May–Leonard competition system. Its first normalized equation is

$$\dot{m}_1 = m_1(1 - m_1 - \alpha m_2 - \beta m_3) + \mu(-2m_1 + m_2 + m_3),$$

with the other two equations cyclically permuted and $\alpha, \beta, \mu > 0$. It has polynomial competition and linear mutation, not the DM receiver-denominator coupling.

Relevant locations. Section IV.B, p. 5, Theorem 1 and Eqs. (7)–(8b): circulant 3×3 Jacobian and explicit spectrum. Section IV.C, pp. 6–7: Theorem 2; general Hopf formula (15), pairing (16), quadratic tensor (18), critical parameter (19), Fourier vectors (22)–(23), and explicit coefficient (24)–(25). Equation (20) and the discussion after (25) exclude the zero-frequency case $\alpha = \beta$. Section IV.A, pp. 4–5 and Fig. 4, and Section IV.E, pp. 10–11, concern folds and other coexistence equilibria; these are not the DM bistability classification.

Overlap and verdict. K: an explicit Fourier-based Hopf calculation in cyclic competition. D/S: a precedent for P03, P11–P13, not an arbitrary-(n, q) DM result. None of the eight exact DM targets was located. The paper itself uses n in a generic normal-form formula on p. 6; that notation does not generalize its three-species model. Its continuation and region descriptions have not been independently validated here.

2.2 S02: Pais, Caicedo-Núñez and Leonard (2012)

Source/version. [11]; supplied `PaisSIADS12.pdf`, author manuscript pp. 1–30. The published article is SIADS 11(4), 1754–1784; the pagination is different.

Exact model and assumptions. Section 2, pp. 3–5, Eq. (2.2): $\dot{x}_i = \sum_j x_j f_j(x) q_{ji} - x_i \phi(x)$, with $f = Bx$, $\phi = x^T Bx$, on the simplex $x_i \geq 0$, $\sum_i x_i = 1$. Equation (2.1), Condition 1 and mutation rules (Q1), (Q2) specify the payoff and mutation restrictions. Section 4, p. 10, Eq. (4.1) uses $B_{C,N} = \text{Circulant}(1, \alpha, 0, \ldots, 0, \beta)$, $\alpha, \beta \in [0, 1)$, $\alpha + \beta > 0$, $N \geq 3$.

Relevant locations. Lemma 4.1, pp. 10–11: mixed equilibrium; Section 4.1, pp. 11–13, Eqs. (4.2)–(4.4), Lemmas 4.2 and 4.4, Theorem 4.5: circulant spectrum, complex pairs, thresholds and crossings. Section 4.2, p. 13, Lemma 4.6: analytical first Lyapunov coefficient. Appendix B, pp. 24–25: general tensor/resolvent formula and normalization. Appendix D, pp. 26–28, Eqs. (D.1)–(D.4): Fourier reduction, including the second-harmonic denominator. Section 5, pp. 16–20, treats cycles with different step sizes; Section 6, pp. 20–23, studies noncirculant examples numerically.

Overlap and verdict. Strong K methodological overlap with P04 and P11–P14: directed cyclic networks, arbitrary dimension and analytical Hopf coefficients are established research territory. The simplex constraint, mutation mechanism and payoff family differ from DM. This is not the canonical arbitrary-(n, q) spectrum, DM quadratic selection theorem or matched DM–MB comparison. Noncirculant simulations do not supply P15–P18's proposed response or robustness theorems.

2.3 S03: van Kekem and Sterk (2018)

Source/version. [13]; supplied `1704.05442v3.pdf`, accepted manuscript dated 21 December 2017, pp. 1–55. Published journal pages are 38–60.

Exact model and assumptions. Section 1.1, p. 1, Eqs. (1a)–(1b):

$$\dot{x}_j = x_{j-1}(x_{j+1} - x_{j-2}) - x_j + F, \qquad x_{j+n} = x_j.$$

The principal first-instability program assumes $n \geq 4$ and increases $F > 0$ from the stable homogeneous state. This quadratic, additively forced system is distinct from both DM and MB.

Relevant locations. Section 2.1, pp. 8–9, Eqs. (3)–(6): Fourier spectrum/vectors; its real symbol uses $f(j, n) = \cos(2\pi j/n) - \cos(4\pi j/n)$. Section 2.2, pp. 10–12: Theorems 2 and 4 and Proposition 5, including simultaneous crossings and the first selected pair. Proposition 5 has a stated $n = 7$ exception and the following remark discusses an $n = 12$ tie. Section 2.3, pp. 12–13, Eq. (8) and Proposition 6: travelling-wave interpretation and the explicit selected-wave-number limit $n/l_1(n) \to 2\pi/\arccos(1/4)$. Section 4.3, pp. 38–46, especially (23)–(25) on pp. 39–40 and (36) on p. 44: general Hopf formula, quadratic reduction and sign analysis. Section 4.4, pp. 47–48, supplies the proof of Proposition 5.

Overlap and verdict. This is the closest core antecedent for targets 2–4 and 6. Quadratic-cosine selection, a discrete first-mode criterion, its large-n limit and a Fourier-reduced Lyapunov coefficient are already present in another cyclic model. A DM coefficient family, all its ties/endpoints and its admissible parameter regions were not located. Their likely category is S rather than a new general Fourier or selection method. The source's positive-F path must not be silently identified with a path in DM's five positive parameters.

2.4 S04: Wei and coauthors (journal issue 2021)

Source/version. [14]; supplied 18-page author manuscript, filename `Wei_et_al_2020_Hopf_bifurcation_and_synchronization.pdf`. The filename's year and 2020 DOI date do not change the 2021 journal issue.

Exact model and assumptions. Section 2, p. 3, Eq. (2.1):

$$\begin{aligned} \dot{x} &= r(y - x) + w, & \dot{y} &= -(1 + m)y + xz - v, \\ \dot{z} &= g[1 + mx^2 - (1 + m)xy], & \dot{w} &= 2(1 + m)w + xz - k_1 x, \\ \dot{v} &= -mv + k_2 y. \end{aligned}$$

Here $r, m, g > 0$ and k_1, k_2 are control parameters. Five is the dimension of a homopolar dynamo, not the number of identical cyclic nodes. It is neither a Wu-Xing coupling model nor DM/MB.

Relevant locations. Section 2, pp. 3–5, Eqs. (2.2)–(2.18), Theorems 2.1–2.2: a Hopf calculation with rounded parameter/eigenvector data. Section 3, pp. 5–7, Eqs. (3.19)–(3.37), Theorems 3.1–3.2: disturbance observer and controlled convergence. The latter assumes the specified control and

observer construction; it is not passive network robustness under small heterogeneity. Section 4, pp. 8–9 and the appended figures supplies control simulations, not a reproduced computation here.

Source cautions. Visually checked p. 3, Eq. (2.6), describes five-component vectors but sums only three components in the pairing; Eq. (2.7) also has inconsistent displayed dimensions. On p. 4, Theorem 2.1 both fixes $m = 0.04$ and lets m pass a different critical value. Theorem 2.2 combines a negative reported coefficient with an unstable-cycle conclusion. These are unresolved manuscript inconsistencies, not corrected theorems in this audit. Equation (2.10)'s coefficient normalization also requires convention matching before comparison.

Overlap and verdict. A generic Hopf/tensor calculation is known; none of the eight exact project targets is located. This paper should not be cited for a cyclic Fourier reduction, a five-element model or DM robustness bounds. Check the final publisher text or an erratum before relying on its detailed criticality claims.

2.5 S05: Lin and Han (2020)

Source/version. [7]; supplied five-page proceedings PDF, printed pp. 777–781. Equations (1)–(2), p. 778, were visually checked because extracted text misreads some subscripts.

Exact model and assumptions. Section 2.1, p. 778, Eq. (1) uses six node-dependent coefficient families and additive interventions e_i. Its homogeneous, intervention-free Eq. (2), written in cyclic notation for comparison, is

$$\dot{z}_i = a_L z_i - b_L z_i^2 + c_L z_i z_{i-1} - d_L z_i z_{i-2} - g_L z_i z_{i+1} - h_L z_i z_{i+2}, \qquad i \in \mathbb{Z}_5.$$

The coefficients are explicitly nonnegative. The exact parameter identification with the canonical five-node MB comparator is

$$(a_L, b_L, c_L, d_L, g_L, h_L) = (a, k, c, h, 0, g).$$

This is a comparison of the displayed equations, not a dynamical theorem. The allowed zero value matters: it makes MB a subfamily, not merely a limiting resemblance.

Relevant locations. Section 2.2, p. 778: positivity discussion; Section 2.2.1, pp. 778–779, Cases 1–4: equilibrium examples, numerical eigenvalues and synchronous scalar reduction. Case 4, p. 779, Eqs. (5)–(7), explicitly treats equal initial activities. Sections 2.2.2–2.2.3, pp. 779–781, and Figs. 3–5 report periodic/chaotic examples and computed Lyapunov exponents. These computations were not rerun or endorsed here. Section 1 and references [9], [10], [14], p. 781, point to earlier Wu-Xing models and perturbation studies.

Overlap and verdict. K: the five-node topology, the six-term polynomial model, its synchronous reduction and $\mathrm{MB}_{5,2}$ as a coefficient subfamily. S/U: the proposed exact comparison between DM and that comparator, not present here. No DM rational law, arbitrary-n spectrum, q=2 all-n selection theorem or Fourier Green function was located. Publication of the five-node topology does not establish the novelty or correctness of the proposed DM analysis.

2.6 S06: Nian and Li (2025)

Source/version. [10]; supplied publisher PDF `s13662-025-03909-9.pdf`, 15 pages, article 61.

Exact model and assumptions. Section 2.2, pp. 5–6, Eq. (1), in the source's order

(M, H, T, S, J), is

$$\dot{M} = a_1 M + b_1 S + c_1 J + d_1 HTSJ,$$
$$\dot{H} = a_2 H + b_2 M + c_2 S + d_2 MTSJ,$$
$$\dot{T} = a_3 T + b_3 H + c_3 M + d_3 MHSJ,$$
$$\dot{S} = a_4 S + b_4 J + c_4 T + d_4 MHTJ,$$
$$\dot{J} = a_5 J + b_5 T + c_5 H + d_5 MHTS.$$

It has node-dependent linear and quartic interactions. Table 1, p. 7, gives parameter meanings; Table 4, p. 12, contains fitted coefficients of mixed signs. The canonical homogeneous positive-parameter domain must not be imposed on this source.

Relevant locations. Section 2.3, pp. 7–8, Eqs. (2)–(5): equilibrium and Jacobian discussion, not a circulant all-n result. Section 3, pp. 8–10, Eqs. (6)–(9): parameter fitting. Section 4, pp. 10–13, Figs. 6–11 and Table 4: illustrative dynamical and Lyapunov-exponent reports. No exact Hopf coefficient or heterogeneous Fourier response theorem is given in these sections.

Overlap and verdict. K: an existing nonlinear five-element application. None of the eight exact mathematical targets was located. It is not DM, not MB and not evidence that five-element dynamics or their numerical bifurcation study is new. Its fitted data, chaos interpretations and displayed Jacobian are not independently validated by this audit and provide no evidence for our model's biological validity.

3 Targeted methodological and five-element supplements

These entries give the closest located general theory or a precise access gap. Bibliographic inclusion does not imply full-text access.

3.1 S07: Gray (2006): circulants and their inverses

[3], Stanford author PDF. Section 3.1, printed pp. 32–34 (PDF positions 40–42), Eqs. (3.7)–(3.13), gives unitary Fourier diagonalization and explicitly notes normality. Section 3.2, Theorem 3.1, printed pp. 34–35 (PDF positions 42–43), includes products and the inverse when all eigenvalues are nonzero. No specific nonlinear network is assumed. These are K general results; the DM spectrum and the finite Fourier inverse kernel are D once its Jacobian and invertibility are verified. Matrix exponentials and resolvents in the same basis are likewise applications of this diagonalization, not a new Fourier method. Gray uses a different Fourier sign convention; indices must be translated. The inverse formula alone supplies no spatial localization estimate.

3.2 S08: Kerin and Engler (2020): general quadratic reduction

[6], arXiv v2, 31-page PDF. Section 3.2, pp. 11–12, Eqs. (27)–(29), Lemma 1: an explicit first Lyapunov coefficient for a real circulant linear part and a quadratic equivariant nonlinearity. Its formula resolves the uniform and second-harmonic interactions through the Laurent symbols. Appendix A, pp. 26–27, Proposition 4 and Eq. (48), supplies the bilinear Fourier interaction rule. Section 5, pp. 21–24, Eq. (40) and Propositions 1–3 treats site-dependent forcing, dissipation and advection, assuming the stated energy-preserving quadratic structure (and positivity of specified diagonal matrices). K: the quadratic Fourier reduction and heterogeneous-network precedents. DM is rational and generally has a nonzero third derivative tensor; its coefficient and sign are not furnished by Lemma 1. Nor are the source's energy hypotheses established for DM.

3.3 S09: Stewart (2024): symmetry and mode ordering

[12], arXiv v1 HTML. Section 3, Theorem 3.7, states equivariant Hopf; Section 4.1, Eqs. (12)–(14), treats a nearest-neighbor unidirectional ring; Section 5.2, Eq. (16), treats longer-range circulant linearizations. Section 5.3, Theorem 5.5 and Corollary 5.6, permits eigenvalue ordering with freely chosen connection strengths. Section 5.4, Theorem 5.7, discusses rotation direction. K: symmetry-based rotating waves and unrestricted circulant mode control. The sign-constrained sparse DM family is not the unrestricted interpolation family. Use its representation-theoretic framework, not genericity as a replacement for P11–P13. The v1 phase displays in Section 5 have apparent indexing inconsistencies; reconcile them against the group action before using a time-shift formula.

3.4 S10: Embree and Trefethen (2001): perturbation bounds

[2], publisher-form author PDF, pp. 583–590. Section 1, pp. 583–584, defines pseudospectra via norm-bounded perturbations and resolvents. Theorem 4_ε, p. 585, gives the similarity/Bauer–Fike inclusion; Theorem 6 and its discussion, p. 586, address normal matrices. K: general matrix perturbation machinery. D: a spectral-margin sufficient bound for an additive matrix perturbation of a stable normal matrix. An explicit bound on changes of DM parameters must additionally control the moving equilibrium and the resulting full Jacobian. This source does not give that parameter certificate or make the heterogeneous Jacobian normal.

3.5 S11: Greenbaum, Li and Overton (2020): sensitivity

[4], 30-page arXiv author PDF. Section 2, p. 4, Assumption 1, Theorem 1 and Eq. (1), gives the derivative of a simple eigenvalue of an analytic matrix family using normalized left and right eigenvectors. Section 3.2, p. 12, states the analytic implicit function theorem used in its argument. K: simple-eigenvalue sensitivity and the implicit function framework. D: first-order threshold sensitivity after the project's parameter family, equilibrium derivative and nonzero crossing derivative are verified. Multiple eigenvalues and mode ties require separate treatment. No DM parameter derivative or explicit robustness radius is supplied.

3.6 S12: Demko, Moss and Smith (1984): localization caution

[1]; partial access only. The publisher-indexed abstract on p. 491 and Section 2, p. 493, Proposition 2.3, Eq. (2.4), and Theorem 2.4 were inspected. They concern decay of inverses of bounded, boundedly invertible band matrices, with spectral/conditioning assumptions and stronger forms for positive definite matrices. K: inverse localization is an established subject. U: the exact reduction to cyclic graph distance, wraparound, arbitrary q and uniform-in-n DM constants. The whole paper and subsequent sparse-matrix extensions require review before any new localization theorem is claimed. A finite Fourier sum is not itself a localization theorem.

3.7 S13: Lin, Shi and Han (2021): unresolved coupling antecedent

[9]. The CCC 2021 proceedings contents identify the title and authors at starting p. 6410; DOI metadata identifies the article. Full text was not available in the audited corpus. Thus its exact coupling equations, assumptions and theorem locations are *unresolved*, not absent. Its enzymatic five-element topic is directly relevant to targets 1 and 5 and the $n = 5, q = 2$ motivation. Retrieve and compare the receiver dependence and any rescaling before claiming the demand-modulated coupling law is unprecedented.

3.8 S14: Lin and Han (2024): integrated five-node overlap risk

[8], publisher landing page, abstract (opening article page 7477) and reference list. The abstract describes enzymatic regulatory networks with $n \leq 9$, oscillation/adaptation comparisons and a five-node topology identified with the yin–yang five-element network. The full equations, analyses and supplement were not accessible. K: the published topic and claimed topology/application overlap; U: exact DM law, analytical spectrum and all eight specific theorem targets. The abstract cannot certify either equation identity or absence. This is a high-priority main-paper literature gap, not postponed archival work.

4 Audit consequences and unresolved evidence

4.1 What the sources do and do not establish

No inspected core paper displays $(\mathrm{DM})_{n,q}$, its receiver-denominator creation term, or the complete DM–MB comparison. That is a finite-corpus finding, not a priority certificate. The five-node bilinear comparator is explicitly contained in S05. General circulant diagonalization/normality, Fourier Hopf reduction, rotating waves, inverse response and matrix/eigenvalue sensitivity all have close antecedents. The detailed classifications for the eight requested targets are in Section 5.

The paper's possible mathematical contribution should therefore be judged by the *verified combination of model-specific conclusions*: the admissible parameter regions, exact mode ties and exclusions, a calibrated bilinear comparison, a nondegenerate Hopf coefficient and parameter-dependent response bounds. This audit does not establish any of them. A change of coupling law does not by itself make every standard calculation a new theorem.

4.2 Correction of the initialization's adjoint concern

S01, p. 6, Eq. (16), S02, Appendix B, pp. 24–25, and S08, p. 12, Eq. (28), use the Hermitian pairing that is conjugate-linear in its first argument, with

$$Jv = \mathrm{i}\omega v, \qquad J^{*}p = -\mathrm{i}\omega p, \qquad \langle p, v\rangle = 1,$$

for a real Jacobian ($J^{*} = J^{T}$). The opposite adjoint sign is a standard consistent convention. The initialization's suggestion that this sign was intrinsically inconsistent is withdrawn in B07 and claim C078. Phase, eigenvector scale, time scaling, Taylor factors and the final DM coefficient still require P13 verification. No master text or proof fragment is changed.

4.3 Priority and access gaps

1. Obtain S13 and S14 with their supplementary equations. Compare the exact nonlinear law, allowable coefficients, variable changes and normalization; abstracts cannot resolve model priority.

2. Follow S05's p. 781 references [9], [10], [14]: T. Wang's 2008 Fudan thesis on the dynamic Wu-Xing model; G. Ding and T. Wang's 2008 mathematical analysis, pp. 269–270; and F. Li's 1987 Yin-yang/Wuxing equations, pp. 966–969. These are *leads quoted from S05's bibliography*, not independently audited sources. Their equations and exact result locations are not asserted here. Resolve them within the main novelty audit.

3. Read the complete S12 paper and follow subsequent inverse-decay and graph-distance results. Determine whether the planned localization estimate is a known specialization.

4. Check the broader cyclic-feedback and structured stability-radius literature once the precise perturbation family, norm and desired bound are fixed. Current matrix theory settles the general mechanism, not the novelty of an unspecified final inequality.

5. Preserve edition-specific citations. The final published Wei text and any corrections have not been compared with the supplied manuscript; source inconsistencies remain disclosed. The Jaramillo article number remains a bibliographic completion item.

4.4 Evidence gates for later work

P04 must verify the actual DM Jacobian before using circulant theory, including coincident offsets. P05 must distinguish a snapshot maximizer from the first crossing on a parameter path; P06 must specify the sequence of coefficients/branches. P08 must fix comparator calibration before any superiority claim. P11–P14 require actual simplicity, nonzero frequency, transversality, nondegeneracy and the correct group action. P15–P18 must distinguish a Fourier inverse at the homogeneous base from the generally noncirculant perturbed problem. These are audit requirements only. All proof and numerical files retain their initial TODO/plan status.

5 Novelty matrix

5.1 Eight targets and source coverage

T1. The exact canonical demand-modulated model and its coupling law.

T2. The arbitrary-(n, q) Fourier spectrum of its homogeneous linearization.

T3. The exact $q = 2$ quadratic mode-selection theorem.

T4. The large-n selected-wave-number limit.

T5. The exact topology- and parameter-count-matched DM–MB comparison.

T6. The explicit first Lyapunov coefficient reduction.

T7. The heterogeneous Fourier Green-function response.

T8. The proposed robustness bounds and Hopf-threshold sensitivity.

Coverage codes (distinct from novelty categories). **G**: an inspected general result supplies the underlying method, conditional on verifying the project's hypotheses; **A**: an inspected analogous calculation in another model; **C**: a component is present (the MB comparator), not the complete comparison; **N**: the exact target was not located in the inspected material; **U**: uninspected content prevents an absence determination. G, A and C do not mean that the source proves the project's target. No row is an exhaustive search of that author's other work.

Source	T1	T2	T3	T4	T5	T6	T7	T8
S01 Jaramillo et al.	N	A	N	N	N	A	N	N
S02 Pais et al.	N	A	N	N	N	A	N	N
S03 van Kekem–Sterk	N	A	A	A	N	A	N	N
S04 Wei et al.	N	N	N	N	N	A	N	A
S05 Lin–Han 2020	N	A	N	N	C	N	N	N
S06 Nian–Li	N	N	N	N	N	N	N	N
S07 Gray	N	G	N	N	N	N	G	G
S08 Kerin–Engler	N	G	A	N	N	G	A	A
S09 Stewart	N	G	A	N	N	N	N	N
S10 Embree–Trefethen	N	N	N	N	N	N	N	G
S11 Greenbaum et al.	N	N	N	N	N	N	G	G
S12 Demko et al. (partial)	U	U	U	U	U	U	G	U
S13 Lin–Shi–Han (lead)	U	U	U	U	U	U	U	U
S14 Lin–Han 2024 (preview)	U	U	U	U	U	U	U	U

S04's T8 analogue is active disturbance-observer control, not a passive spectral stability radius. S05's T2 analogue is a numerical five-node linearization example, not an arbitrary-n theorem. S08's T6 general formula assumes a quadratic nonlinearity; that assumption does not cover the entire rational DM field. S11 supplies implicit-function and eigenvalue-sensitivity ingredients, not a Fourier kernel. S12 supplies inverse-decay theory in the inspected excerpt; its full cyclic application remains U. These qualifications are part of the matrix.

5.2 Claim-level novelty decisions

The K/D/S/T/U categories are defined in Section 1. A mixed entry separates a known mechanism from an unestablished model-specific contribution. Every D judgment is conditional; no hypotheses are proved in Phase 1.

Target / owner	Classification	Evidence and precise boundary
T1 00; P01–P03	S, with U priority gap	The exact DM equation was not located in the six supplied models. S05, p. 778, Eqs. (1)–(2), is bilinear; S06, p. 6, Eq. (1), is linear/quartic. S13's full model and S14's full enzymatic equations are unavailable. The tentative contribution is the particular coupling-law combination and its analysis. Neither five-element topology nor rational regulation in general may be called new. No source inspected justifies calling the creation term a standard Holling type-II response.
T2 P03–P04	K general theory; D DM application	S07, Theorem 3.1, printed pp. 34–35, already covers every circulant size and coefficient pattern. S02, p. 11, Eqs. (4.2)–(4.3), and S03, p. 9, Eqs. (3)–(6), apply it to networks. DM derivative coefficients, Fourier-sign translation and coincident offsets remain unverified. An exact DM formula can be a useful lemma, but Fourier diagonalization or normality is not the novelty.
T3 P05	S; D quadratic mechanism	S03, p. 9, Eq. (4), and Proposition 5, pp. 11–12, select modes of a two-harmonic cosine symbol and account for a simultaneous first crossing. Section 4.4 gives its proof. The prospective DM addition is a complete classification over its admissible coefficient region, including endpoints, ties and uniform/alternating modes. The reviewed paper does not state that exact classification. A general quadratic maximization method should not be claimed as new.
T4 P06	D for a fixed limiting symbol; U for stronger scope	S03, Proposition 6, p. 13, already gives an explicit large-n selected-wave-number limit from its continuous symbol maximum. A DM limit under fixed coefficients and a unique limiting maximizer is an analogous specialization. A varying-parameter family, ties, uniform rates or a first-Hopf limit requires a precise statement and additional literature review. A limiting spectral distribution alone would not establish convergence of maximizers.
T5 P08	K $MB_{5,2}$ component; S exact comparison	S05, p. 778, Eq. (2), contains $MB_{5,2}$ with the explicit coefficient map in Section 2.5. Its synchronous reduction is already in Case 4, p. 779, Eqs. (5)–(7). No exact DM–MB comparison was located. Same topology and parameter count do not fix a meaningful calibration. General-n comparison regions may be a contribution after P08 and the missing coupling antecedents are resolved.

Target / owner	Classification	Evidence and precise boundary
T6 P11–P13	K Fourier reduction; S DM formula; U sign theorem	S02, Lemma 4.6 and Appendix D; S03, Theorem 4 and Eq. (36); S08, Lemma 1, Eq. (29), already reduce Hopf coefficients using Fourier mode interactions. S01, Eqs. (15)–(25), is a three-node instance. The rational DM tensor contractions and a rigorously nonzero sign on a stated parameter region have not been supplied by these sources or calculated here. Generic reduction is not a new theorem; a proved model-specific sign region could be.
T7 P15–P17	D finite base-state kernel; U localization novelty	S07, Theorem 3.1, supplies a circulant inverse; S11, Section 3.2, supplies an implicit-function framework. Thus Fourier inversion of the first-order homogeneous-base sensitivity is known machinery. It is not an exact diagonalization after arbitrary heterogeneity. S12, p. 493, Proposition 2.3 and Theorem 2.4, prevents treating inverse localization itself as unprecedented. Cyclic distance, directional structure, uniform constants and allowable q remain to be specified and compared.
T8 P18	D matrix bound and simple threshold derivative; U explicit parameter theorem	S10, Theorem 4_ε, p. 585, and normal-matrix discussion, p. 586; S11, Theorem 1, p. 4: standard perturbation bounds and simple-eigenvalue derivatives. A sufficient DM parameter radius must include equilibrium movement, positive coefficients and denominator control. A threshold shift also needs a nonzero crossing derivative. The final norm, uncertainty class and constants are not yet fixed; theorem-level novelty is unresolved.

5.3 Five-node specialization within the main program

The motivating $n = 5, q = 2$ case should appear as a worked specialization of the arbitrary-(n, q) results. S05 provides the exact bilinear antecedent; S06 provides a different five-element polynomial application; S13–S14 are immediate coupling-law and topology-priority checks. S04's five dimensions do not make it a five-element network antecedent. S09, Section 4.2 and Section 5.4, already discusses five-node phase patterns.

The fifth-root/golden-ratio evaluations in P07 are D special-function algebra after P04–P05. Any sharp parameter inequalities obtained from them would be S unless an additional conclusion warrants a different classification. A fifth-root expression or the occurrence of rotating waves on five nodes is not, by itself, a defensible novelty claim.

5.4 Manuscript wording gates

Claim that must not yet be written	Gate and acceptable audit conclusion
"First Fourier spectrum for cyclic networks"	Known general theory, S07; cite it. P04 must verify the specific model.

Claim that must not yet be written	Gate and acceptable audit conclusion
"First exact quadratic mode-selection or large-n wave-number result"	S03 is a direct precedent. Specify the additional DM parameter/tie classification and complete P05–P06.
"New five-element bilinear model"	$MB_{5,2}$ is contained in S05. Describe it as the matched comparator, and resolve S13–S14 before claiming a new coupling law.
"First closed Fourier formula for a Hopf coefficient"	S02, S03 and S08 already contain such reductions. P13 must deliver the DM expression and any claimed sign certificate.
"Fourier Green function proves localization under heterogeneity"	P15–P16 must state the expansion and error bound; S12 and related graph-distance literature must be checked for localization.
"Robustness follows from the homogeneous spectral gap"	P18 must bound the actual changed Jacobian at the moved equilibrium in the stated norm. A matrix bound is not yet a parameter radius.
"Synchronous bistability implies bistability of the whole network"	P10 requires full-system stability, with basin scope explicit. None of the source analogues establishes this claim for DM.
"A generic rotating-wave theorem proves our Hopf branch is stable"	P11–P14 must verify all spectral/nondegeneracy hypotheses and stability. Generic symmetry results do not supply those facts.

Release decision. Phase 1 provides a citable, version-specific literature assessment and an explicit queue of unresolved evidence. It does not authorize a "first," "novel theorem" or quantitative superiority assertion. No separate archival study is needed to settle these main-program gaps.

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